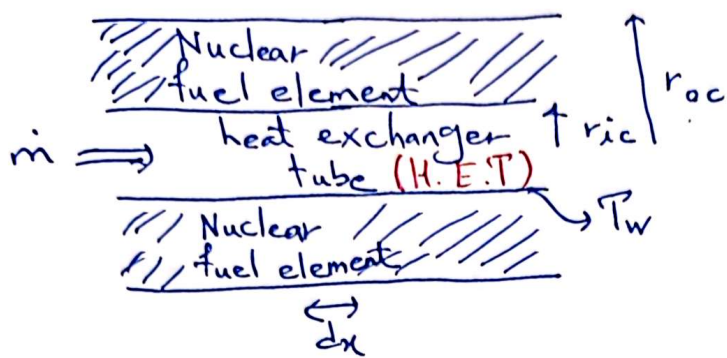




$$Q = P_s \nabla$$

Q : Heat absorbed in tube $= \dot{m} q = \dot{m} c_p dT_o$

P_s : Power density of nuclear reaction

∇ : Volume of nuclear reaction (nuclear energy production)



front view

$$\nabla = \underbrace{\pi (r_{oc}^2 - r_{ic}^2)}_{\text{Area}} \underbrace{dx}_{\text{length}}$$

$$\Rightarrow \dot{m} c_p dT_o = P_s \pi (r_{oc}^2 - r_{ic}^2) dx \Rightarrow \boxed{\frac{dT_o}{dx} = \frac{P_s \pi (r_{oc}^2 - r_{ic}^2)}{\dot{m} c_p}}$$

if $P_s = \text{cte} \rightarrow \frac{dT_o}{dx} = \text{cte}$

on the other side, Q in the H.E.T is ~~equal~~ ^{due to} convection heat transfer

$$\underbrace{\dot{m} c_p dT_o}_{Q} = \underbrace{Q}_{\text{convection}} = \underbrace{h}_{\substack{\text{convection heat} \\ \text{transfer coef.}}} \underbrace{(T_w - T_o)}_{\substack{\text{wall temp.} \\ \text{tube perimeter} \\ \text{flow temp}}} \underbrace{cdx}_{\text{tube perimeter}}$$

We can rewrite the last Eqⁿ.

$$\frac{dT_o}{dx} = \frac{h(T_w - T_o)c}{\dot{m}c_p}$$

Now, we use Reynolds analogy: (analogy between heat transfer & fluid dynamics), it says:

$$\begin{aligned} \text{Convection Coef.} \leftarrow \frac{h}{\dot{m}c_p} &= \frac{f}{2A_E} \rightarrow \text{Friction coef} \\ &= \frac{f}{2\pi \frac{D^2}{4}} = \frac{2f}{\pi D^2} \end{aligned}$$

$$\frac{dT_o}{dx} = \frac{2f}{\pi D^2} (T_w - T_o) \cancel{\pi D^2} = \frac{2f}{D} (T_w - T_o)$$

remember $P_s = dT_o \rightarrow \frac{dT_o}{dx} = dT_o$ so $(T_w - T_o) = dT_o$

$$\Rightarrow \frac{dT_o}{dx} = \frac{2f}{D} (T_w - T_o) = \frac{2f}{D} (T_{w2} - T_{o2})$$

2: heat exchanger exit

To maximize $T_{o2} \rightarrow T_{w2}$ should also be maximized
(so that the difference be maximized)

$$\begin{aligned} \Rightarrow \frac{dT_o}{dx} &= \frac{2f}{D} (T_{w_{max}} - T_{o2}) \quad \int_{T_{o1}}^{T_{o2}} = \int_0^L \\ \Rightarrow T_{o2} - T_{o1} &= \frac{2f}{D} (T_{w_{max}} - T_{o2}) L \Rightarrow T_{o2} = \frac{T_{o1} + \frac{2fL}{D} T_{w_{max}}}{1 + \frac{2fL}{D}} \end{aligned}$$